Beyond Broad Applications: Can Pathology Foundation Models Adapt to Hematopathology?

Abstract. Recent advances in foundation models offer promising, scalable solutions for digital pathology, especially for classification tasks with limited data. However, most studies treat these models as static feature extractors, overlooking their full potential in real-world scenarios. Meanwhile, accurate blood cell morphology analysis is vital for diagnosing disorders like anemia and leukemia, particularly in resource-limited settings. In this study, we systematically evaluate the robustness and adaptability of four pathology foundation models specifically Gigapath, PLIP, DinoBloom, and PhikonV2, across five hematology datasets with diverse sizes, class distributions, and clinical contexts. We benchmark each foundation model under different training paradigms and compare them against standard deep learning architectures. Our results reveal that model rankings are highly dependent on the size of the training data and model configuration, with foundation models often underperforming or matching simpler models when not fine-tuned appropriately. These findings emphasize the need for robust, multi-regime benchmarking to fairly assess foundation models in pathology, especially in domain-specific, low-data applications like hematology.

Keywords: Foundation Models · Hematology · Digital Pathology · Fine-Tuning · Whole-Slide Images · Vision Transformers · Benchmarking.

1 Introduction

Pathologists around the world face mounting pressures as demand increasingly outweighs supply, creating critical shortages in healthcare systems [16, 19]. The modern pathologist's role has expanded beyond traditional microscopic examination to encompass complex molecular assays and genetic testing, further straining this limited workforce. Simultaneously, the digitization of pathology slides has created unprecedented opportunities for computational approaches, with artificial intelligence (AI) emerging as a promising solution to these challenges [10, 18]. Foundation models (FM), large-scale AI systems designed to adapt across diverse applications, are now transforming computational pathology through their scalable architectures and multimodal integration capabilities, with particularly significant implications for specialized fields like hematopathology [8, 6].

The evolution of AI in pathology has progressed from narrow task-specific tools to sophisticated FMs capable of addressing the unique challenges of pathology data. These challenges range from processing gigapixel whole-slide images (WSIs) to analyzing single-cell morphology, while enabling knowledge transfer across institutions and disease domains. Unlike conventional AI approaches that require extensive task-specific training, FMs can adapt to applications they weren't specifically designed for, offering remarkable flexibility in clinical and research settings. This adaptability is particularly valuable in hematopathology, where morphological subtleties and rare cellular events demand both contextual understanding and precise cellular analysis.

Microsoft's GigaPath [21] represents a paradigm shift in computational pathology with its innovative "LongNet" architecture, which processes entire WSIs as continuous sequences rather than fragmented patches. Trained on more than 170,000 slides from 28 US cancer centers, GigaPath's 2024 release included hematology-specific benchmarks that achieved 94% AUC in predicting TP53 mutations from acute myeloid leukemia slides, a 12% improvement over previous patch-based methods. The model's global context modeling capability has proven especially valuable for capturing rare events like circulating tumor cells in peripheral blood smears, a task where traditional convolutional neural networks (CNNs) often fail due to their limited receptive fields.

For multimodal integration, PLIP (Pathology Language and Image Pretraining) [12] provides an accessible framework through its HuggingFace implementation. It was trained using billion of pathology image–text pairs from Twitter and other entities including LAION. Its lightweight design facilitates deployment on edge devices in resource-limited hematology clinics, addressing practical implementation concerns.

Phikon-v2 [9] addresses scalability challenges through self-supervised pretraining on 460 million pathology tiles. A key distinctive feature that make Phikon-v2 notable is that it was trained to encode granular morphological features such as nuclear shape, tissue texture, and cellular patterns which are critical in hematopathology and other fine-detailed domains.

While general pathology FMs offer broad tissue analysis capabilities, DinoBloom [13] stands apart as the first specialized FM for hematology, representing a significant advancement in single-cell analysis. Unlike general pathology FMs that struggle with cellular-level resolution, DinoBloom was purposefully designed for hematologic applications and trained on 380,000 white blood cell images across diverse preparation methods. The 2025 extension of DinoBloom to minimal residual disease (MRD) detection in acute lymphoblastic leukemia further highlights its specialized advantage, demonstrating 0.1% sensitivity that surpasses both conventional flow cytometry thresholds and the detection limits of general pathology FMs when applied to hematologic specimens.

However, despite the potential of these FMs, their application in classification tasks within hematopathology remains underexplored. A key challenge lies in their limited adaptation to the domain-specific nuances of hematopathological imagery. FMs are typically trained on broad, heterogeneous datasets, which may not capture the fine-grained morphological features critical for accurate diagnosis in this field. This raises concerns about their reliability and clinical relevance when deployed in specialized settings without further tuning or domain-aware evaluation.

In this research, we investigate the adaptability of pathology FMs to hematopathology specific classification tasks. By leveraging pre-trained FMs and fine-tuning them on specialized hematopathology datasets, this paper seeks to fill the gap by evaluating several FMs through rigorous external testing across multiple datasets, providing a framework for responsible clinical translation. The results highlight both the remarkable potential and practical limitations of FMs in specialized pathology domains, offering insights into their optimal implementation for improving diagnostic accuracy, reproducibility, and efficiency in hematopathology practice.

2 Experimental Setup and Methods

This study evaluated the performance of four FMs (GigaPath, PLIP, PhikonV2, and Dinobloom) across different experimental settings and datasets in hematopathology classification tasks. The models were trained and evaluated using 5-fold cross-validation, with performance assessed using three key metrics: accuracy, F1-score, and the area under the curve (AUC). These results were then compared against the performance of three state-of-the-art deep learning (DL) models: MobileNetV3 [14], ResNet50 [11], and the Vision Transformer (ViT) [7] which has become a standard architecture due to its excellent performance on many computer vision benchmarks [4, 15]. Each model was evaluated under two distinct training paradigms: (1) trained from scratch with randomly initialized weights, and (2) used as a feature extractor by loading pretrained weights from ImageNet [17] and fine-tuning only the final classification layer while keeping all other layers frozen. This dual setting allows for a comprehensive assessment of both model capacity and transfer learning effectiveness in the context of limited medical imaging data.

2.1 Pipeline for the Classification Task

We evaluated the performance of the FMs across three distinct experimental settings:

- **Feature Extraction without Fine-Tuning (Setting 1):** The FMs were employed solely as fixed feature extractors, without any additional training or fine-tuning on the target dataset. The embeddings generated by the pretrained FMs served as input features for downstream classification.
- **Full Data Fine-Tuning of the Last Layer (Setting 2):** We fine-tuned only the final classification layer of the FMs using the entire available training set, except for a reserved test set comprising 100 images per class. This setting aims to assess the model's adaptability when trained on a comprehensive dataset while preserving a balanced and unseen evaluation subset.
- **Partial Data Fine-Tuning of the Last Layer (Setting 3):** Similar to setting 2, we fine-tuned only the last layer; however, the training data was reduced to half of the total available samples, again holding out 100 images per class for testing. This condition simulates limited data availability and evaluates the robustness of fine-tuning under constrained data regimes.

2.2 Evaluation Datasets

In this study, we comprehensively evaluate binary and multi-class classification performance across five diverse digital hematopathology datasets, which include both multi-cell and single-cell images, to ensure robust and generalizable conclusions in computational pathology. ALL image dataset (D1) [3] includes 3,256 peripheral blood smear images from 89 patients at Taleqani Hospital (Iran), categorized into benign hematogones and malignant lymphoblast subtypes, captured at 100x magnification (224×224 JPG). The WBC dataset (D2) [22] comprises 300 white blood cell images (120×120 pixels) from Jiangxi Tecom, China, stained with rapid hematology reagent and showing yellowish backgrounds. The ALL-IDB2 dataset (D3) [1] contains 280 cropped images focusing on individual cells (normal vs malignant) and is widely used for leukemia-related algorithm development. The PBS Barcelona (D4) [2] offers 17,092 expert-annotated images of normal blood cells in eight classes, acquired at the Hospital Clinic of Barcelona using a CellaVision DM96 analyzer (360×363 JPG). Finally, the Bodzas dataset (D5) includes 16,027 high-resolution annotated images of 9 white blood cell types, derived from 12,986 samples of 78 patients (leukemic and non-leukemic), collected between 2020 and 2022 with a resolution of 42 pixels /µm. Table 1 provides a detailed overview of these datasets.

Table 1: Summary of the datasets used in this study.

ID	Images	Classes	Class Names
D1	3,256	4	Benign, Early Pre-B, Pre-B, Pro-B ALL
D2	300	4	Neutrophil, Lymphocyte, Monocyte, Eosinophil
D3	260	2	Normal, Malignant
D4	17,101	8	Neutrophils, Eosinophils, Basophils, Lymphocytes, Monocytes, Immature granulocytes, Erythroblasts, Platelets
D5	16,037	9	Neutrophil segments, Neutrophil bands, Eosinophils, Basophils, Lymphocytes, Monocytes, Normoblasts, Myeloblasts

3 Results and Discussion

3.1 Evaluation

Initially, we evaluated four FMs for histopathology using a zero-shot approach (Setting 1), wherein the models generated embeddings that were directly used for classification without any task-specific fine-tuning. Experiments were conducted on diagnostic images from five distinct datasets to assess model generalizability. Figure 1 presents the accuracy, F1-score, and AUC for each FM across all datasets.

Dinobloom demonstrates the best performance by achieving the highest accuracy, AUC and F1-score across D1, D2, D3 and D4 compared with the general pathology FMs. On the D1, we noted the performance was not significantly different from GigaPath. We also found that the performance of all models has decreased on D2 with dinobloom achieving substantial improvement against the other FMs.

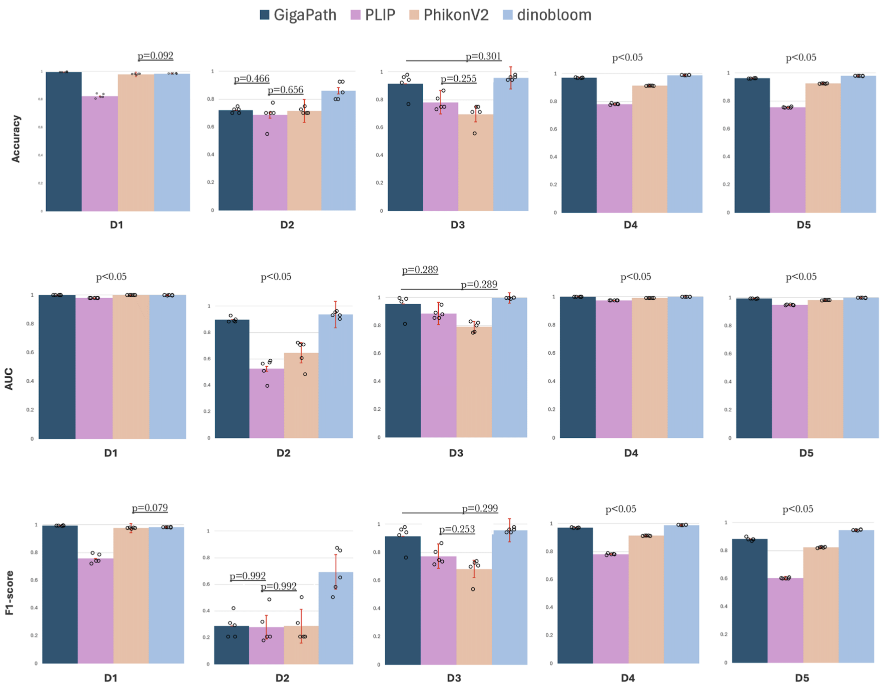

Fig. 1: Performance comparison of GigaPath, PLIP, PhikonV2 and dinobloom across five datasets using a zero-shot approach. Bar plots illustrate accuracy, F1-score, and AUC for each model on each dataset. The error bars show the standard error across 5 independent experiments. The p-values indicate the significance level using paired t-tests with FDR correction

Figure 2 presents t-SNE (t-Distributed Stochastic Neighbor Embedding) [20] visualizations of the image embeddings generated by the four FMs across the five datasets. These plots offer qualitative support for the quantitative performance metrics reported in Figure 1. In particular, embeddings from Dataset D2 consistently show less clear class separation across all models, highlighting the difficulty in distinguishing its classes. Across all datasets, DinoBloom consistently produces embeddings with clearer cluster separation, characterized by distinct and well-formed clusters. This comparative visualization highlights DinoBloom's ability to generate more discriminative representations than the other models, regardless of the dataset.

3.2 Foundation models under different settings

To study the impact of fine-tuning on the performance of the FMs, we conducted several comparison under the settings described in 2.1. Due to dataset size limitations, only datasets D1 and D4 were included in this evaluation. Other datasets lacked sufficient image samples per class to reliably support the described experimental protocols, particularly the fixed-size test splits.

In Setting 1 with the D1 dataset, GigaPath demonstrated exceptional performance with an accuracy of 0.9942 ($\pm$ 0.0017), F1-score of 0.9937 ($\pm$ 0.0019), and near-perfect AUC of 0.9999 ($\pm$ 0.0001). This performance can be attributed to its innovative "LongNet" architecture, which processes entire whole-slide images as continuous sequences rather than fragmented patches. This approach allows GigaPath to capture global contextual information that might be missed by models with more limited receptive fields. Dinobloom followed closely with an accuracy of 0.9834 ($\pm$ 0.0023), F1-score of 0.9826 ($\pm$ 0.0026), and AUC of 0.9986 ($\pm$ 0.0002). In Setting 2 with the D1 dataset, all models demonstrated improved consistency in their performance. Dinobloom achieved the highest accuracy at 0.99 ($\pm$ 0.006) and F1-score at 0.990 ($\pm$ 0.006), with a perfect AUC of 1.000 ($\pm$ 0.000). GigaPath, PhikonV2, and PLIP followed with accuracies of 0.982 ($\pm$ 0.007), 0.984 ($\pm$ 0.005), and 0.960 ($\pm$ 0.025), respectively. The F1-scores mirrored these results closely, and all models achieved AUC values of 0.999 or higher, indicating excellent discriminative ability. Setting 3 with D1 revealed slightly more variability in model performance with GigaPath maintaining the strongest results with an accuracy of 0.966 ($\pm$ 0.023) and F1-score of 0.965 ($\pm$ 0.024). Dataset D4 revealed Dinobloom as the top performer with an accuracy and F1-score surpassing 0.98 and perfect AUC.

The consistently high performance of Dinobloom, pretrained on hematology data exclusively, across different settings and datasets demonstrates the potential of specialized FMs in hematopathology. Dinobloom appears to provide a significant advantage in hematopathology classification tasks. This aligns with previous findings that FMs with domain-specific pretraining can outperform more general models when applied to specialized tasks [5].

PhikonV2 showed consistent performance across all settings and datasets, while PLIP achieved the lowest relative performance among the four models. This performance gap may be attributed to its design focus on multimodal integration rather than pure image classification. PLIP's architecture, which links cytomorphological features to clinical text entries, may not be optimally aligned with the pure classification tasks evaluated in this study. Nevertheless, its improved performance in Settings 2 and 3 suggests that PLIP can adapt effectively to specific hematopathology tasks when provided with appropriate experimental conditions. Moreover, the consistently low performance across all models on certain datasets (D2 and D3) highlights the critical impact of dataset size on model effectiveness.

3.3 DL-based Prediction Performance

Figure 3 reveals a comprehensive performance comparison of three DL models (MobileNetV3, ResNet50, and ViT) across the five datasets (D1-D5) using Accuracy, F1-score, and AUC metrics, demonstrating that datasets D1, D4, and D5 consistently yield higher performance while D2 and D3 present significant challenges, particularly for F1-score.

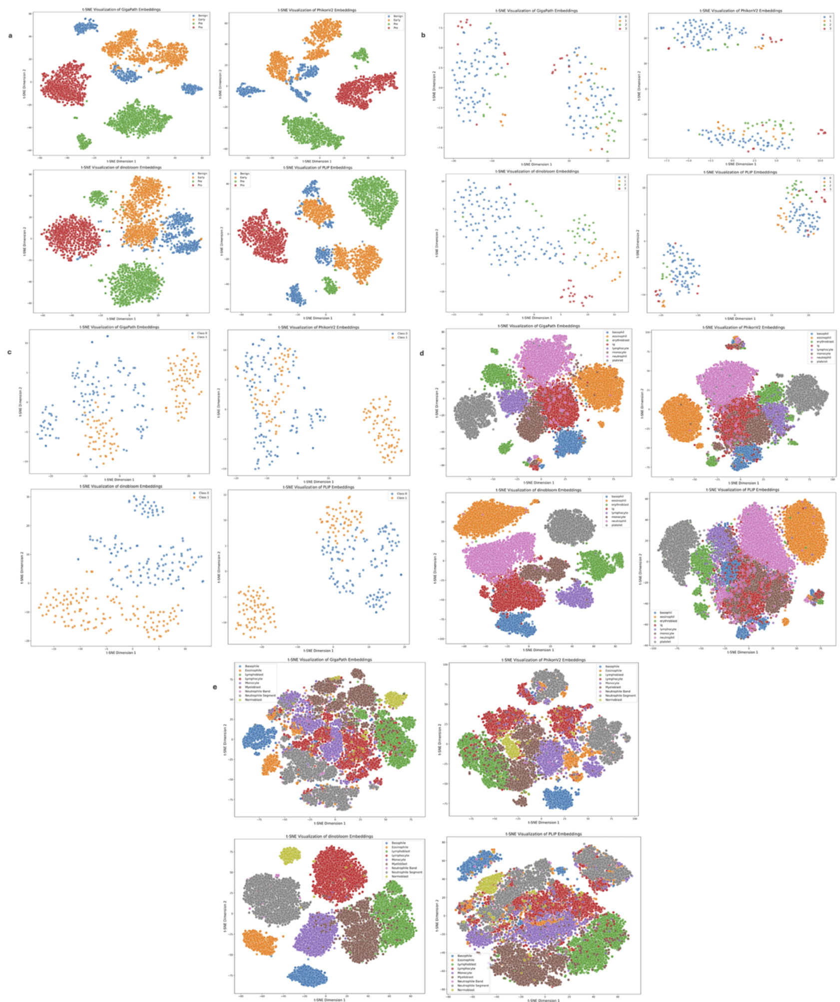

Fig. 2: t-SNE visualizations of embeddings from GigaPath, PLIP, PhikonV2 and dinobloom across datasets D1-D5 (a-e). Each plot displays two-dimensional projections of high-dimensional feature spaces, where each point represents an image and colors indicate different classes.

Fine-tuning substantially improves performance across all models and datasets with MobileNetV3 showing the most dramatic improvements (especially for D3 where Accuracy increased from 0.519 to 0.746 and F1-score from 0.342 to 0.744); ResNet50 emerges as the most consistent performer across datasets with moderate improvements from fine-tuning, while ViT occupies an intermediate position with better generalization than MobileNetV3 but less consistency than ResNet50.

Experimental results on DL models revealed a significant finding: FMs may not be necessary for this particular task. Well-tuned, lightweight DL models can achieve performance comparable to larger FMs, offering a more practical and efficient option for clinical deployment in hematologic image classification.

4 Conclusion

This study provides a thorough evaluation of four foundation models: GigaPath, PLIP, PhikonV2, and Dinobloom, for hematopathology classification across multiple datasets and experimental setups. The findings highlight key performance trends, strengths, and limitations of each model within the specialized context of digital pathology. Among them, Dinobloom consistently delivered superior results, with several configurations achieving accuracy, F1-score, and AUC values above 0.95. Its strong performance is likely attributed to an architecture well-suited for cellular-level image analysis. The observed variability across models and configurations highlights the importance of thoughtful model selection, dataset curation, and configuration tuning in clinical applications. Future work should explore the versatility of these models across a broader range of clinically relevant tasks and incorporate a wider array of deep learning-based models, particularly those specifically designed for medical image classification, to fully assess their utility in real-world medical imaging workflows. Overall, foundation models show significant promise for advancing hematopathology diagnostics, offering potential solutions to workforce shortages and rising diagnostic complexity when appropriately validated and deployed.

Table 2: Model performance across three settings for Data1 and Data4 using Accuracy, F1-score, and AUC (mean ± std). Best values per setting and metric are in bold.

		D1			**D4**		
Setting	**Model**	Accuracy	F1-score	AUC	Accuracy	F1-score	AUC
Setting 1	GigaPath	**0.9942 ± 0.0017**	**0.9937 ± 0.0019**	**0.9999 ± 0.0001**	0.970 ± 0.002	0.969 ± 0.003	0.999 ± 0.000
	PLIP	0.8227 ± 0.0157	0.7568 ± 0.0331	0.9794 ± 0.0008	0.781 ± 0.005	0.721 ± 0.009	0.975 ± 0.000
	PhikonV2	0.9782 ± 0.0033	0.9767 ± 0.0032	0.9992 ± 0.0001	0.913 ± 0.002	0.900 ± 0.002	0.992 ± 0.000
	Dinobloom	0.9834 ± 0.0023	0.9826 ± 0.0026	0.9986 ± 0.0002	**0.989 ± 0.001**	**0.990 ± 0.001**	**1.000 ± 0.000**
Setting 2	GigaPath	0.982 ± 0.007	0.981 ± 0.007	**1.000 ± 0.000**	0.9862 ± 0.0046	0.9863 ± 0.0046	0.9998 ± 0.0002
	PLIP	0.960 ± 0.025	0.959 ± 0.025	0.999 ± 0.000	0.9757 ± 0.0077	0.9758 ± 0.0076	0.9995 ± 0.0003
	PhikonV2	0.984 ± 0.005	0.984 ± 0.005	0.999 ± 0.000	0.9867 ± 0.0037	0.9868 ± 0.0037	**0.9998 ± 0.0001**
	Dinobloom	**0.990 ± 0.006**	**0.990 ± 0.006**	**1.000 ± 0.000**	**0.9908 ± 0.0036**	**0.9908 ± 0.0036**	**0.9998 ± 0.0001**
Setting 3	GigaPath	**0.966 ± 0.023**	**0.965 ± 0.024**	**1.000 ± 0.001**	0.988 ± 0.003	0.988 ± 0.003	**1.000 ± 0.000**
	PLIP	0.919 ± 0.025	0.916 ± 0.027	0.996 ± 0.002	0.976 ± 0.005	0.976 ± 0.005	0.999 ± 0.000
	PhikonV2	0.947 ± 0.007	0.946 ± 0.008	0.996 ± 0.002	0.982 ± 0.004	0.982 ± 0.004	**1.000 ± 0.000**
	Dinobloom	0.938 ± 0.025	0.937 ± 0.026	0.989 ± 0.007	**0.991 ± 0.001**	**0.991 ± 0.001**	**1.000 ± 0.000**

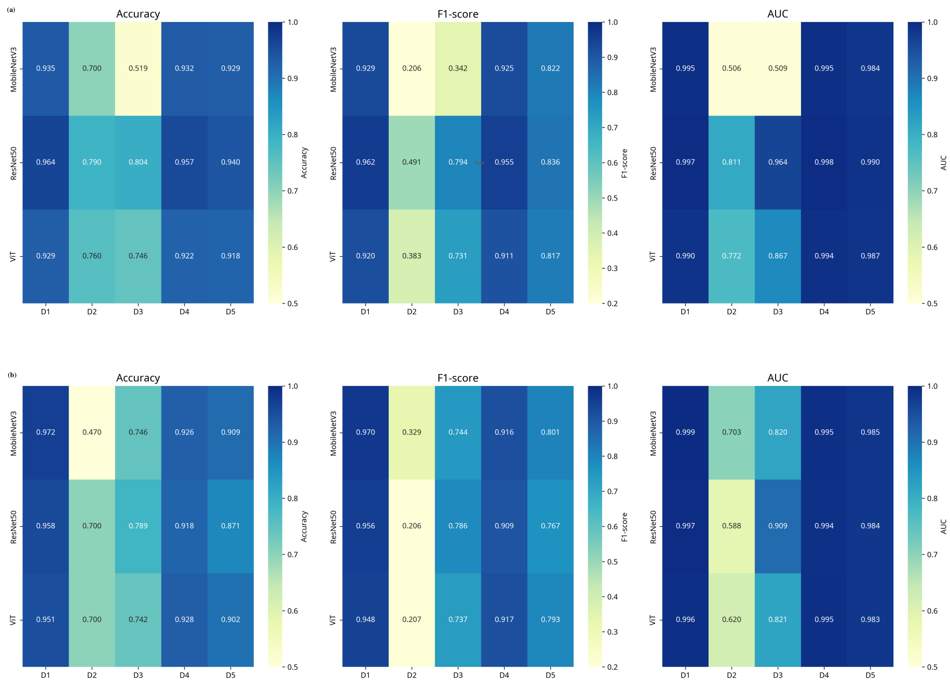

Fig. 3: Performance of the DL models (a)without fine-tuning (b)with fine-tuning.

Disclosure of Interests. The authors declare no conflicts of interest.

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